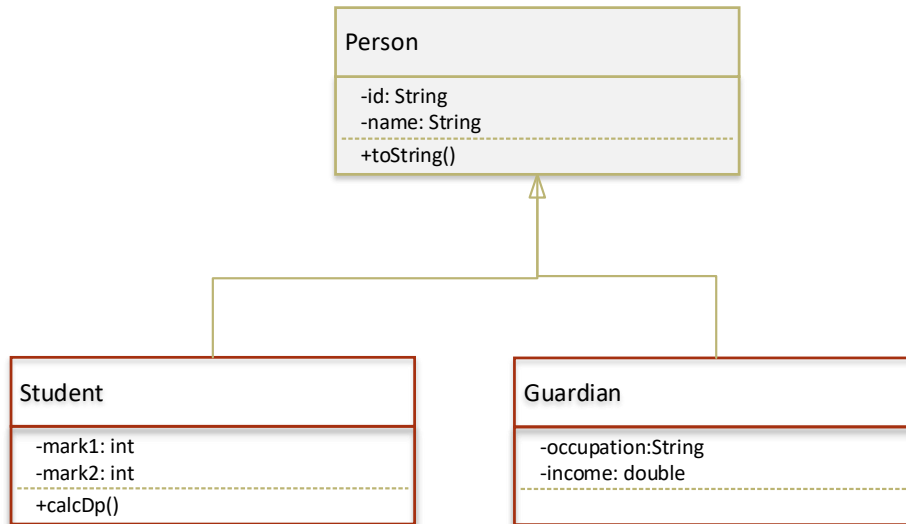

Topic 1 Exercise 1: Inheritance (2 marks for completion)



NOTE: Use ONLY **Person** and **Student** classes in the UML diagram for this exercise

1. Create a package **inherit**.
2. In this package:
 - a. Write the **Person** and **Student** classes (separately). Each class should contain **at least two constructors, get and set, and toString() methods**. **As explained in the lecture, re-use code (constructors, get and set, toString()) defined in the super class to ensure there is no duplication in Student class.**
 - b. Write a **Tester** (driver) which calls parameterized constructors to create and display one instance of **Person** (called **p1**) and **Student** (called **s1**).
 - c. Run the application
3. Modify the **Tester** (by adding statements to do the following) and run the program
 - a. Create another instance of **Person** (called **p**).
 - b. Assign **s1** to **p** (**p = s1**) and display **p**.
 - Can you explain your answer?
4. Modify **Tester** as follows:
 - a. Assign **p** to **s1** (i.e., **s1=p**). Can you explain the error?
 - b. How can you fix the error? **Solution: type casting**